10 minutes to pandas
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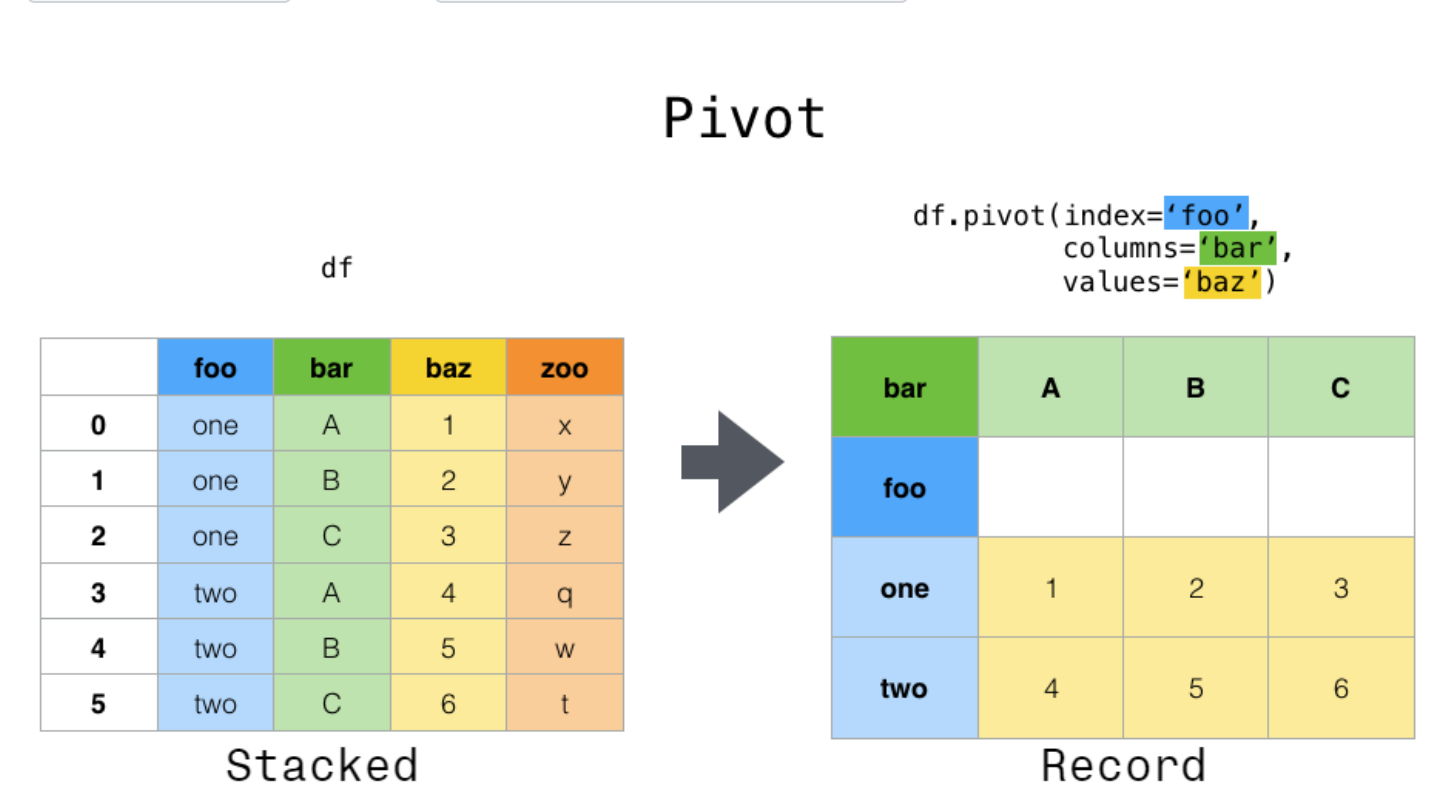
`explode()`
`crosstab()`
`cut()`
`factorize()`

Reshaping and pivot tables

pandas provides methods for manipulating a `Series` and `DataFrame` to alter the representation of the data for further data processing or data summarization.

- `pivot()` and `pivot_table()`: Group unique values within one or more discrete categories.
- `stack()` and `unstack()`: Pivot a column or row level to the opposite axis respectively.
- `melt()` and `wide_to_long()`: Unpivot a wide `DataFrame` to a long format.
- `get_dummies()` and `from_dummies()`: Conversions with indicator variables.
- `explode()`: Convert a column of list-like values to individual rows.
- `crosstab()`: Calculate a cross-tabulation of multiple 1 dimensional factor arrays.
- `cut()`: Transform continuous variables to discrete, categorical values
- `factorize()`: Encode 1 dimensional variables into integer labels.

`pivot()` and `pivot_table()`



`pivot()`

Data is often stored in so-called "stacked" or "record" format. In a "record" or "wide" format, typically there is one row for each subject. In the "stacked" or "long" format there are multiple rows for each subject where applicable.

```
In [1]: data = {
   ...:     "value": range(12),
   ...:     "variable": ["A"] * 3 + ["B"] * 3 + ["C"] * 3 + ["D"] * 3,
   ...:     "date": pd.to_datetime(["2020-01-03", "2020-01-04", "2020-01-05"] * 4)
   ...: }
   ...:

In [2]: df = pd.DataFrame(data)
```

To perform time series operations with each unique variable, a better representation would be where the `columns` are the unique variables and an `index` of dates identifies individual observations. To reshape the data into this form, we use the `DataFrame.pivot()` method (also implemented as a top level function `pivot()`):

```
In [3]: pivoted = df.pivot(index="date", columns="variable", values="value")

In [4]: pivoted
Out[4]:
variable    A  B  C   D
date
2020-01-03  0  3  6   9
2020-01-04  1  4  7  10
2020-01-05  2  5  8  11
```

If the `values` argument is omitted, and the input `DataFrame` has more than one column of values which are not used as column or index inputs to `pivot()`, then the resulting "pivoted" `DataFrame` will have hierarchical columns whose topmost level indicates the respective value column:

```
In [5]: df["value2"] = df["value"] * 2

In [6]: pivoted = df.pivot(index="date", columns="variable")

In [7]: pivoted
Out[7]:
           value           value2
variable       A  B  C   D      A   B   C   D
date
2020-01-03     0  3  6   9      0   6  12  18
2020-01-04     1  4  7  10      2   8  14  20
2020-01-05     2  5  8  11      4  10  16  22
```

You can then select subsets from the pivoted `DataFrame`:

```
In [8]: pivoted["value2"]
Out[8]:
variable    A   B   C   D
date
2020-01-03  0   6  12  18
2020-01-04  2   8  14  20
2020-01-05  4  10  16  22
```

Note that this returns a view on the underlying data in the case where the data are homogeneously-typed.

> **Note**
>
> `pivot()` can only handle unique rows specified by `index` and `columns`. If you data contains duplicates, use `pivot_table()`.

`pivot_table()`

While `pivot()` provides general purpose pivoting with various data types, pandas also provides `pivot_table()` or `pivot_table()` for pivoting with aggregation of numeric data.

The function `pivot_table()` can be used to create spreadsheet-style pivot tables. See the cookbook for some advanced strategies.

```
In [9]: import datetime

In [10]: df = pd.DataFrame(
   ....:     {
   ....:         "A": ["one", "one", "two", "three"] * 6,
   ....:         "B": ["A", "B", "C"] * 8,
   ....:         "C": ["foo", "foo", "foo", "bar", "bar", "bar"] * 4,
   ....:         "D": np.random.randn(24),
   ....:         "E": np.random.randn(24),
   ....:         "F": [datetime.datetime(2013, i, 1) for i in range(1, 13)]
   ....:         + [datetime.datetime(2013, i, 15) for i in range(1, 13)],
   ....:     }
   ....: )
   ....:

In [11]: df
Out[11]:
        A  B    C         D         E          F
0     one  A  foo  0.469112  0.404705 2013-01-01
1     one  B  foo -0.282863  0.577046 2013-02-01
2     two  C  foo -1.509059 -1.715002 2013-03-01
3   three  A  bar -1.135632 -1.039268 2013-04-01
4     one  B  bar  1.212112 -0.370647 2013-05-01
..    ... ..  ...       ...       ...        ...
19  three  B  foo -1.087401 -0.472035 2013-08-15
20    one  C  foo -0.673690 -0.013960 2013-09-15
21    one  A  bar  0.113648 -0.362543 2013-10-15
22    two  B  bar -1.478427 -0.006154 2013-11-15
23  three  C  bar  0.524988 -0.923061 2013-12-15

[24 rows x 6 columns]

In [12]: pd.pivot_table(df, values="D", index=["A", "B"], columns=["C"])
Out[12]:
C             bar       foo
A     B
one   A -0.995460  0.595334
      B  0.393570 -0.494817
      C  0.196903 -0.767769
three A -0.431886       NaN
      B       NaN -1.065818
      C  0.798396       NaN
two   A       NaN  0.197720
      B -0.986678       NaN
      C       NaN -1.274317

In [13]: pd.pivot_table(
   ....:     df, values=["D", "E"],
   ....:     index=["B"],
   ....:     columns=["A", "C"],
   ....:     aggfunc="sum",
   ....: )
   ....:
Out[13]:
          D                          ...         E
A       one                three     ...     three       two
C       bar       foo        bar     ...       foo       bar       foo
B                                    ...
A -1.990921  1.190667 -0.863772  ...       NaN       NaN -1.067650
B  0.787140 -0.989634       NaN  ...  0.372851  1.63741       NaN
C  0.393806 -1.535539  1.596791  ...       NaN       NaN -3.491906

[3 rows x 12 columns]

In [14]: pd.pivot_table(
   ....:     df, values="E",
   ....:     index=["B", "C"],
   ....:     columns=["A"],
   ....:     aggfunc=["sum", "mean"],
   ....: )
   ....:
Out[14]:
             sum                                mean
A            one     three       two        one     three       two
B C
A bar -0.471593 -2.008182       NaN -0.235796 -1.004091       NaN
  foo  0.761726       NaN -1.067650  0.380863       NaN -0.533825
B bar -1.665170       NaN  1.637410 -0.832585       NaN  0.818705
  foo -0.097554  0.372851       NaN -0.048777  0.186425       NaN
C bar -0.744154 -2.392449       NaN -0.372077 -1.196224       NaN
  foo  1.061810       NaN -3.491906  0.530905       NaN -1.745953
```

The result is a `DataFrame` potentially having a `MultiIndex` on the index or column. If the `values` column name is not given, the pivot table will include all of the data in an additional level of hierarchy in the columns:

```
In [15]: pd.pivot_table(df[["A", "B", "C", "D", "E"]], index=["A", "B"], columns=["
Out[15]:
                D                   E
C             bar       foo       bar       foo
A     B
one   A -0.995460  0.595334 -0.235796  0.380863
      B  0.393570 -0.494817 -0.832585 -0.048777
      C  0.196903 -0.767769 -0.372077  0.530905
three A -0.431886       NaN -1.004091       NaN
      B       NaN -1.065818       NaN  0.186425
      C  0.798396       NaN -1.196224       NaN
two   A       NaN  0.197720       NaN -0.533825
      B -0.986678       NaN  0.818705       NaN
      C       NaN -1.274317       NaN -1.745953
```

Also, you can use `Grouper` for `index` and `columns` keywords. For detail of `Grouper`, see Grouping with a Grouper specification.

```
In [16]: pd.pivot_table(df, values="D", index=pd.Grouper(freq="ME", key="F"), colum
Out[16]:
C                bar       foo
F
2013-01-31       NaN  0.595334
2013-02-28       NaN -0.494817
2013-03-31       NaN -1.274317
2013-04-30 -0.431886       NaN
2013-05-31  0.393570       NaN
2013-06-30  0.196903       NaN
2013-07-31       NaN  0.197720
2013-08-31       NaN -1.065818
2013-09-30       NaN -0.767769
2013-10-31 -0.995460       NaN
2013-11-30 -0.986678       NaN
2013-12-31  0.798396       NaN
```

Adding margins

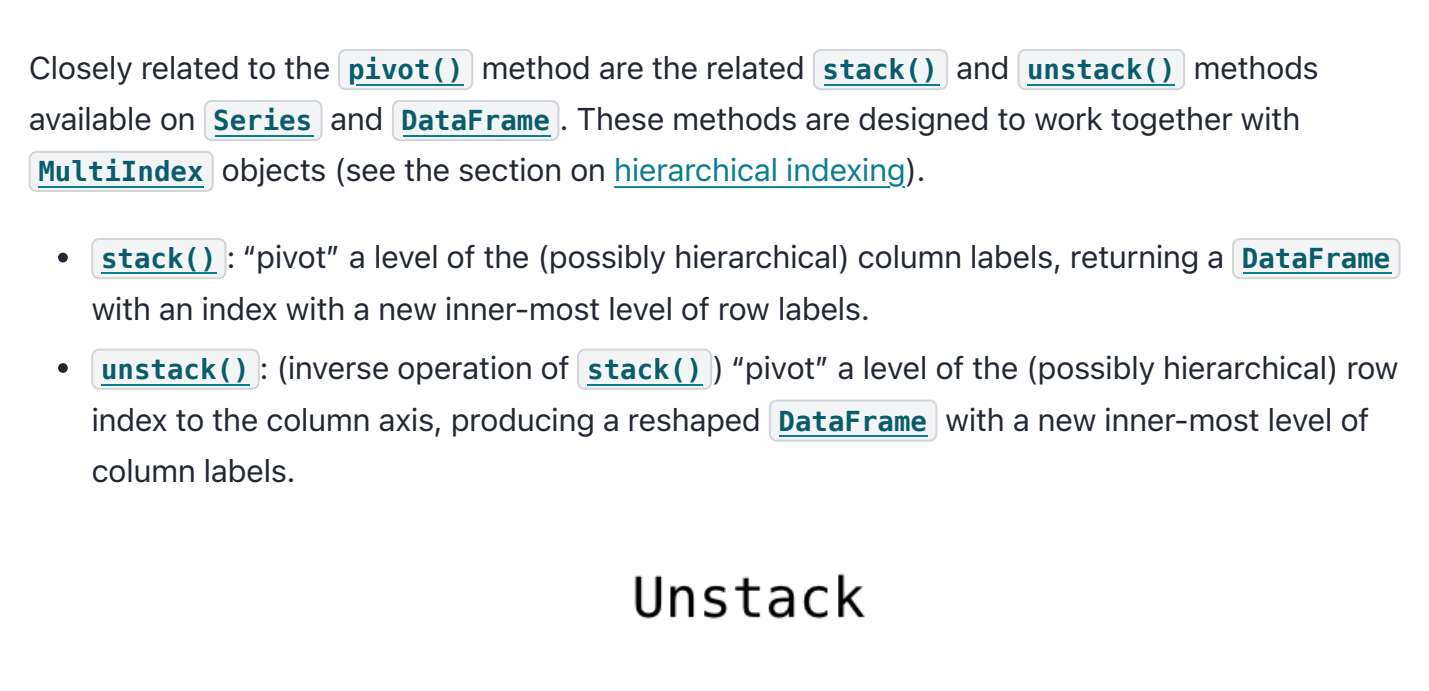
Passing `margins=True` to `pivot_table()` will add a row and column with an `All` label with partial group aggregates across the categories on the rows and columns:

```
In [17]: table = df.pivot_table(
   ....:     index=["A", "B"],
   ....:     columns="C",
   ....:     values=["D", "E"],
   ....:     margins=True,
   ....:     aggfunc="std"
   ....: )
   ....:

In [18]: table
Out[18]:
                D                             E
C             bar       foo       All       bar       foo       All
A     B
one   A  1.568517  0.178504  1.293926  0.179247  0.033718  0.371275
      B  1.157593  0.299748  0.860059  0.653280  0.885047  0.779837
      C  0.523425  0.133049  0.638297  1.111310  0.770555  0.938819
three A  0.995247       NaN  0.995247  0.049748       NaN  0.049748
      B       NaN  0.030522  0.030522       NaN  0.931203  0.931203
      C  0.386657       NaN  0.386657  0.386312       NaN  0.386312
two   A       NaN  0.111032  0.111032       NaN  1.146201  1.146201
      B  0.695438       NaN  0.695438  1.166526       NaN  1.166526
      C       NaN  0.331975  0.331975       NaN  0.043771  0.043771
All      1.014073  0.713941  0.871016  0.881376  0.984017  0.923568
```

Additionally, you can call `DataFrame.stack()` to display a pivoted DataFrame as having a multi-level index:

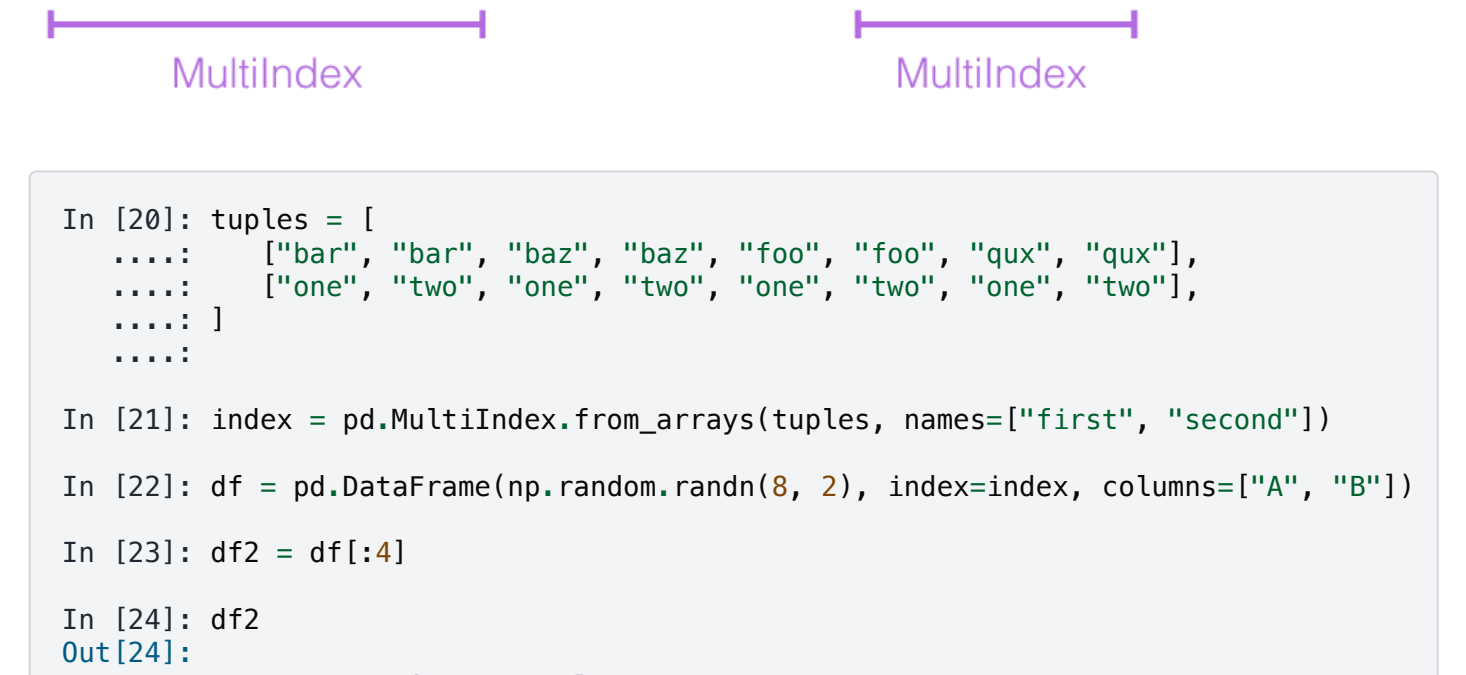
```
In [19]: table.stack()
Out[19]:
                  D         E
A   B C
one A bar  1.568517  0.179247
      foo  0.178504  0.033718
      All  1.293926  0.371275
    B bar  1.157593  0.653280
      foo  0.299748  0.885047
...             ...       ...
two C foo  0.331975  0.043771
      All  0.331975  0.043771
All   bar  1.014073  0.881376
      foo  0.713941  0.984017
      All  0.871016  0.923568

[30 rows x 2 columns]
```

`stack()` and `unstack()`



Closely related to the `pivot()` method are the related `stack()` and `unstack()` methods available on `Series` and `DataFrame`. These methods are designed to work together with `MultiIndex` objects (see the section on hierarchical indexing).

- `stack()`: "pivot" a level of the (possibly hierarchical) column labels, returning a `DataFrame` with an index with a new inner-most level of row labels.
- `unstack()`: (inverse operation of `stack()`) "pivot" a level of the (possibly hierarchical) row index to the column axis, producing a reshaped `DataFrame` with a new inner-most level of column labels.



```
In [20]: tuples = [
   ....:    ["bar", "bar", "baz", "baz", "foo", "foo", "qux", "qux"],
   ....:    ["one", "two", "one", "two", "one", "two", "one", "two"],
   ....: ]
   ....:

In [21]: index = pd.MultiIndex.from_arrays(tuples, names=["first", "second"])

In [22]: df = pd.DataFrame(np.random.randn(8, 2), index=index, columns=["A", "B"])

In [23]: df2 = df[:4]

In [24]: df2
Out[24]:
                     A         B
first second
bar   one     0.895717  0.805244
      two    -1.206412  2.565646
baz   one     1.431256  1.340309
      two    -1.170299 -0.226169
```

The `stack()` function "compresses" a level in the `DataFrame` columns to produce either:

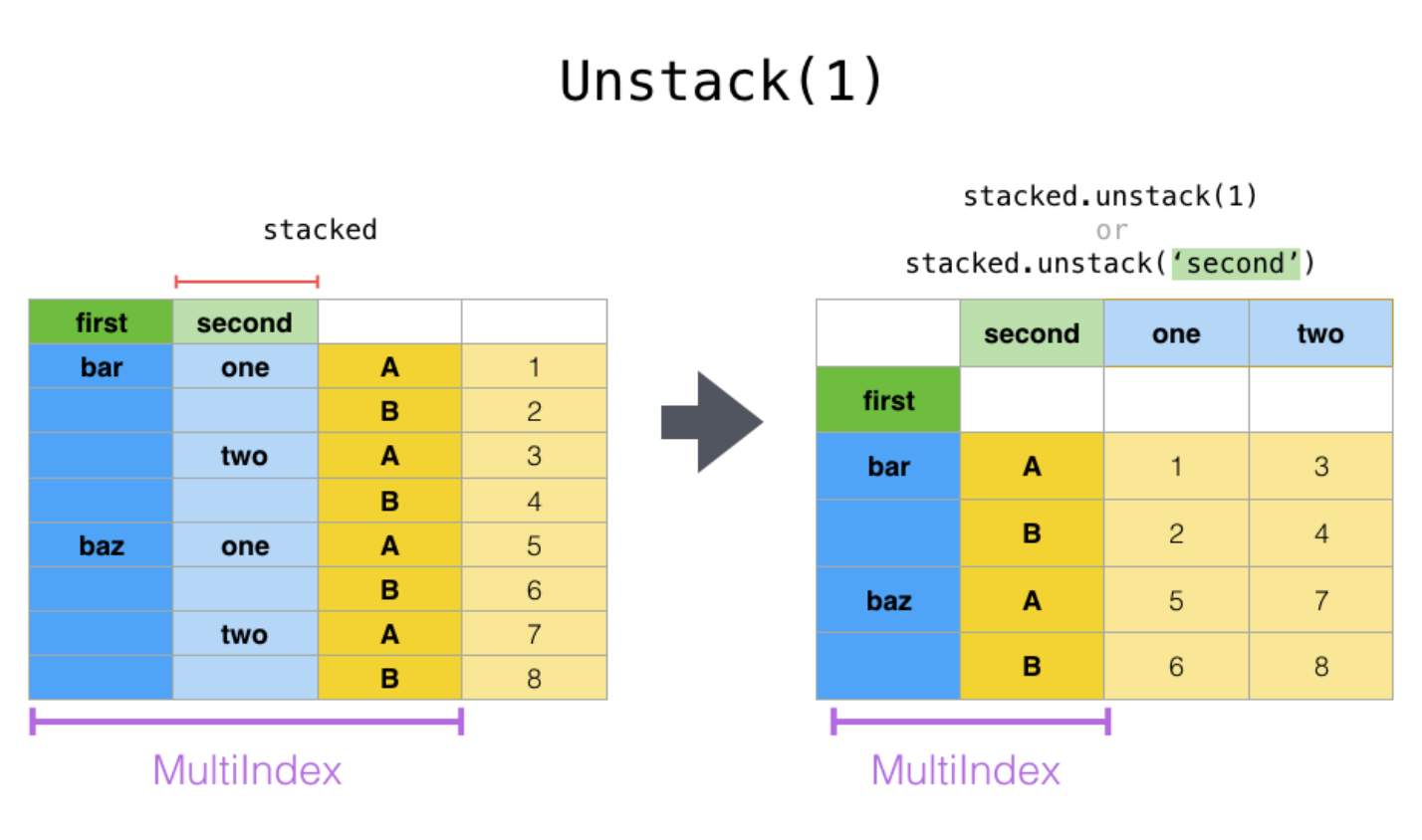
- A `Series`, in the case of a `Index` in the columns.
- A `DataFrame`, in the case of a `MultiIndex` in the columns.

If the columns have a `MultiIndex`, you can choose which level to stack. The stacked level becomes the new lowest level in a `MultiIndex` on the columns:

```
In [25]: stacked = df2.stack()

In [26]: stacked
Out[26]:
first  second
bar    one     A    0.895717
               B    0.805244
       two     A   -1.206412
               B    2.565646
baz    one     A    1.431256
               B    1.340309
       two     A   -1.170299
               B   -0.226169
dtype: float64
```

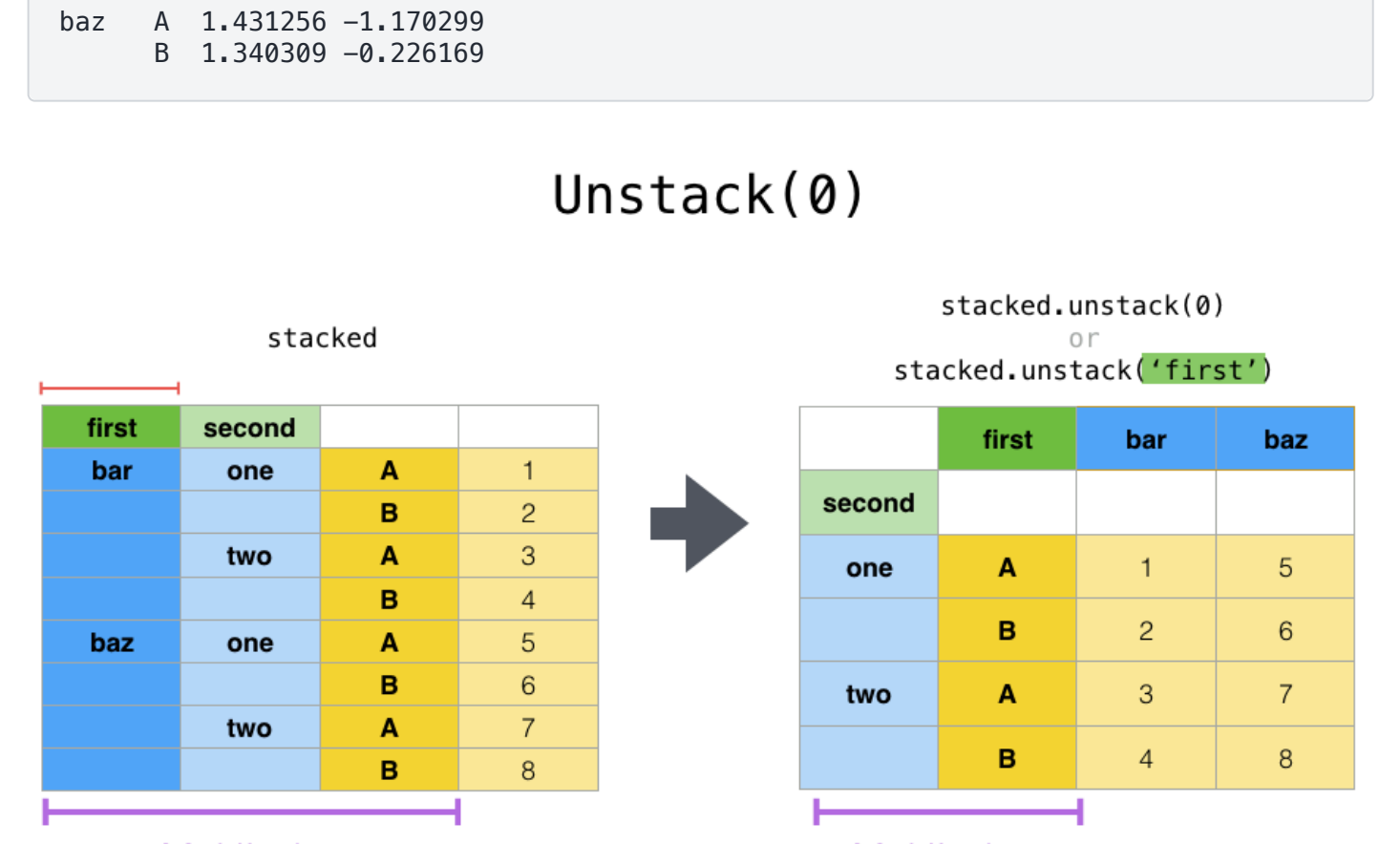
With a "stacked" `DataFrame` or `Series` (having a `MultiIndex` as the `index`), the inverse operation of `stack()` is `unstack()`, which by default unstacks the **last level**:

```
In [27]: stacked.unstack()
Out[27]:
                     A         B
first second
bar   one     0.895717  0.805244
      two    -1.206412  2.565646
baz   one     1.431256  1.340309
      two    -1.170299 -0.226169

In [28]: stacked.unstack(1)
Out[28]:
second        one       two
first
bar   A  0.895717 -1.206412
      B  0.805244  2.565646
baz   A  1.431256 -1.170299
      B  1.340309 -0.226169

In [29]: stacked.unstack(0)
Out[29]:
first          bar       baz
second
one    A  0.895717  1.431256
       B  0.805244  1.340309
two    A -1.206412 -1.170299
       B  2.565646 -0.226169
```



If the indexes have names, you can use the level names instead of specifying the level numbers:

```
In [30]: stacked.unstack("second")
Out[30]:
second        one       two
first
bar   A  0.895717 -1.206412
      B  0.805244  2.565646
baz   A  1.431256 -1.170299
      B  1.340309 -0.226169
```



Notice that the `stack()` and `unstack()` methods implicitly sort the index levels involved. Hence a call to `stack()` and then `unstack()`, or vice versa, will result in a **sorted** copy of the original `DataFrame` or `Series`:

```
In [31]: index = pd.MultiIndex.from_product([[2, 1], ["a", "b"]])

In [32]: df = pd.DataFrame(np.random.randn(4), index=index, columns=["A"])

In [33]: df
Out[33]:
            A
2 a -1.413681
  b  1.607920
1 a  1.024180
  b  0.569605

In [34]: all(df.unstack().stack() == df.sort_index())
Out[34]: True
```

Multiple levels

You may also stack or unstack more than one level at a time by passing a list of levels, in which case the end result is as if each level in the list were processed individually.

```
In [35]: columns = pd.MultiIndex.from_tuples(
   ....:     [
   ....:         ("A", "cat", "long"),
   ....:         ("B", "cat", "long"),
   ....:         ("A", "dog", "short"),
   ....:         ("B", "dog", "short"),
   ....:     ],
   ....:     names=["exp", "animal", "hair_length"],
   ....: )
   ....:

In [36]: df = pd.DataFrame(np.random.randn(4, 4), columns=columns)

In [37]: df
Out[37]:
exp                 A         B         A         B
animal            cat       cat       dog       dog
hair_length      long      long     short     short
0            0.875906 -2.211372  0.974466 -2.006747
1           -0.410001 -0.078638  0.545952 -1.219217
2           -1.226825  0.769804 -1.281247 -0.727707
3           -0.121306 -0.097883  0.695775  0.341734

In [38]: df.stack(level=["animal", "hair_length"])
Out[38]:
exp                          A         B
  animal hair_length
0 cat    long         0.875906 -2.211372
  dog    short        0.974466 -2.006747
1 cat    long        -0.410001 -0.078638
  dog    short        0.545952 -1.219217
2 cat    long        -1.226825  0.769804
  dog    short       -1.281247 -0.727707
3 cat    long        -0.121306 -0.097883
  dog    short        0.695775  0.341734
```

The list of levels can contain either level names or level numbers but not a mixture of the two.

```
# df.stack(level=['animal', 'hair_length'])
# from above is equivalent to:
In [39]: df.stack(level=[1, 2])
Out[39]:
exp                          A         B
  animal hair_length
0 cat    long         0.875906 -2.211372
  dog    short        0.974466 -2.006747
1 cat    long        -0.410001 -0.078638
  dog    short        0.545952 -1.219217
2 cat    long        -1.226825  0.769804
  dog    short       -1.281247 -0.727707
3 cat    long        -0.121306 -0.097883
  dog    short        0.695775  0.341734
```

Missing data

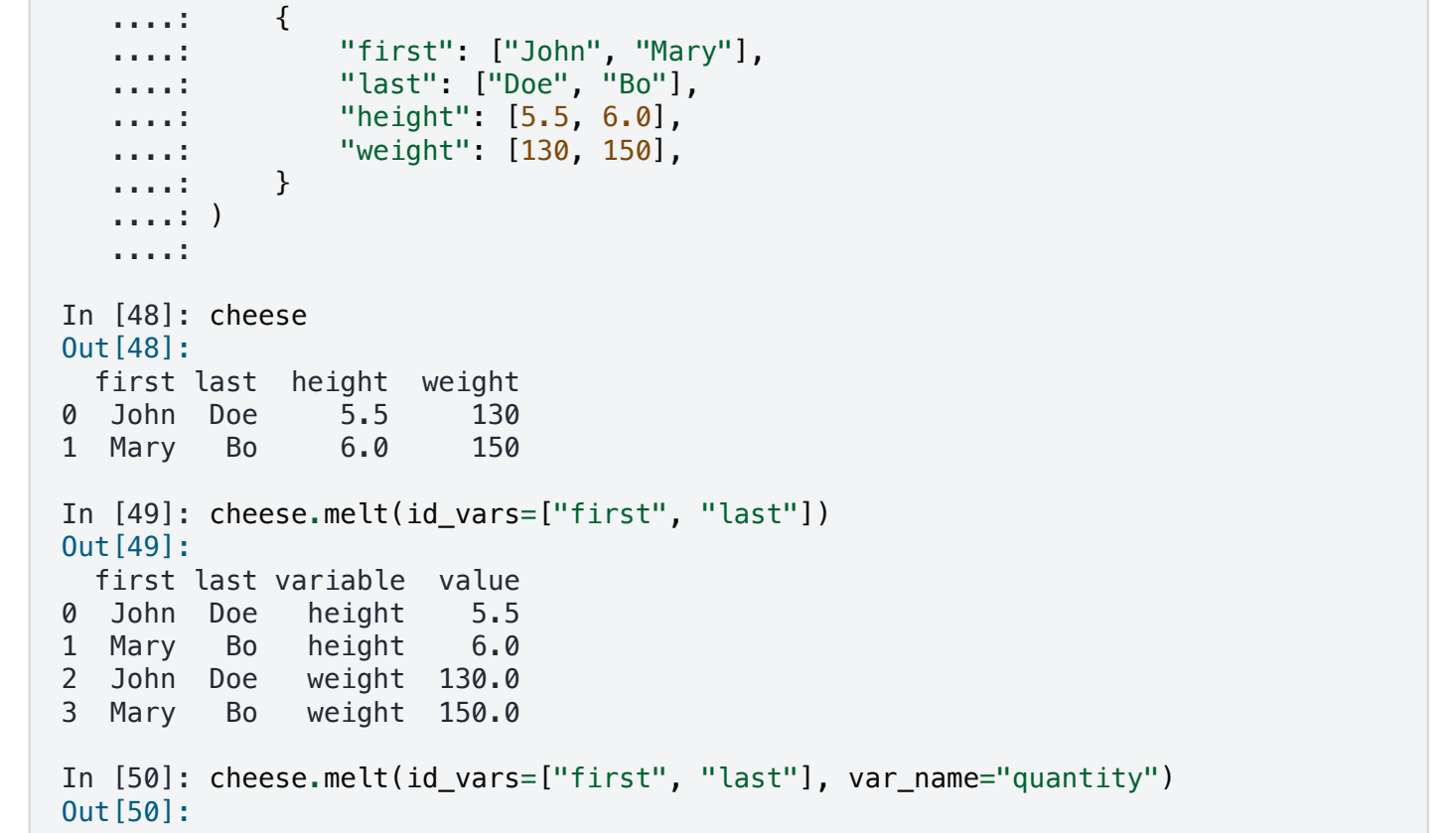
Unstacking can result in missing values if subgroups do not have the same set of labels. By default, missing values will be replaced with the default fill value for that data type.

```
In [40]: columns = pd.MultiIndex.from_tuples(
   ....:     [
   ....:         ("A", "cat"),
   ....:         ("B", "dog"),
   ....:         ("B", "cat"),
   ....:         ("A", "dog"),
   ....:     ],
   ....:     names=["exp", "animal"],
   ....: )
   ....:

In [41]: index = pd.MultiIndex.from_product(
   ....:     [("bar", "baz", "foo", "qux"), ("one", "two")], names=["first", "secon
   ....: )
   ....:

In [42]: df = pd.DataFrame(np.random.randn(8, 4), index=index, columns=columns)

In [43]: df3 = df.iloc[[0, 1, 4, 7], [1, 2]]

In [44]: df3
Out[44]:
exp                  B
animal             dog       cat
first second
bar   one    -1.110336 -0.619976
      two     0.687738  0.176444
foo   one     1.314232  0.690579
qux   two     0.380396  0.084844

In [45]: df3.unstack()
Out[45]:
exp           B
animal      dog                 cat
second      one       two       one       two
first
bar   -1.110336  0.687738 -0.619976  0.176444
foo    1.314232       NaN  0.690579       NaN
qux         NaN  0.380396       NaN  0.084844
```

The missing value can be filled with a specific value with the `fill_value` argument.

```
In [46]: df3.unstack(fill_value=-1e9)
Out[46]:
exp                B
animal           dog                           cat
second           one           two             one           two
first
bar    -1.110336e+00  6.877384e-01 -6.199759e-01  1.764443e-01
foo     1.314232e+00 -1.000000e+09  6.905793e-01 -1.000000e+09
qux    -1.000000e+09  3.803956e-01 -1.000000e+09  8.484421e-02
```

`melt()` and `wide_to_long()`



The top-level `melt()` function and the corresponding `DataFrame.melt()` are useful to reshape a `DataFrame` into a format where one or more columns are *identifier variables*, while all other columns, considered *measured variables*, are "unpivoted" to the row axis, leaving just two non-identifier columns, "variable" and "value". The names of those columns can be customized by supplying the `var_name` and `value_name` parameters.

```
In [47]: cheese = pd.DataFrame(
   ....:     {
   ....:         "first": ["John", "Mary"],
   ....:         "last": ["Doe", "Bo"],
   ....:         "height": [5.5, 6.0],
   ....:         "weight": [130, 150],
   ....:     }
   ....: )
   ....:

In [48]: cheese
Out[48]:
  first last  height  weight
0  John  Doe     5.5     130
1  Mary   Bo     6.0     150

In [49]: cheese.melt(id_vars=["first", "last"])
Out[49]:
  first last variable  value
0  John  Doe   height    5.5
1  Mary   Bo   height    6.0
2  John  Doe   weight  130.0
3  Mary   Bo   weight  150.0

In [50]: cheese.melt(id_vars=["first", "last"], var_name="quantity")
Out[50]:
  first last quantity  value
0  John  Doe   height    5.5
1  Mary   Bo   height    6.0
2  John  Doe   weight  130.0
3  Mary   Bo   weight  150.0
```

When transforming a DataFrame using `melt()`, the index will be ignored. The original index values can be kept by setting the `ignore_index` parameter to `False` (default is `True`). `ignore_index=False` will however duplicate index values.

```
In [51]: index = pd.MultiIndex.from_tuples([("person", "A"), ("person", "B")])

In [52]: cheese = pd.DataFrame(
   ....:     {
   ....:         "first": ["John", "Mary"],
   ....:         "last": ["Doe", "Bo"],
   ....:         "height": [5.5, 6.0],
   ....:         "weight": [130, 150],
   ....:     },
   ....:     index=index,
   ....: )
   ....:

In [53]: cheese
Out[53]:
         first last  height  weight
person A  John  Doe     5.5     130
       B  Mary   Bo     6.0     150

In [54]: cheese.melt(id_vars=["first", "last"])
Out[54]:
  first last variable  value
0  John  Doe   height    5.5
1  Mary   Bo   height    6.0
2  John  Doe   weight  130.0
3  Mary   Bo   weight  150.0

In [55]: cheese.melt(id_vars=["first", "last"], ignore_index=False)
Out[55]:
         first last variable  value
person A  John  Doe   height    5.5
       B  Mary   Bo   height    6.0
       A  John  Doe   weight  130.0
       B  Mary   Bo   weight  150.0
```

`wide_to_long()` is similar to `melt()` with more customization for column matching.

```
In [56]: dft = pd.DataFrame(
   ....:     {
   ....:         "A1970": {0: "a", 1: "b", 2: "c"},
   ....:         "A1980": {0: "d", 1: "e", 2: "f"},
   ....:         "B1970": {0: 2.5, 1: 1.2, 2: 0.7},
   ....:         "B1980": {0: 3.2, 1: 1.3, 2: 0.1},
   ....:         "X": dict(zip(range(3), np.random.randn(3))),
   ....:     }
   ....: )
   ....:

In [57]: dft["id"] = dft.index

In [58]: dft
Out[58]:
  A1970 A1980  B1970  B1980         X  id
0     a     d    2.5    3.2  1.519970   0
1     b     e    1.2    1.3 -0.493662   1
2     c     f    0.7    0.1  0.600178   2

In [59]: pd.wide_to_long(dft, ["A", "B"], i="id", j="year")
Out[59]:
                X  A    B
id year
0  1970  1.519970  a  2.5
1  1970 -0.493662  b  1.2
2  1970  0.600178  c  0.7
0  1980  1.519970  d  3.2
1  1980 -0.493662  e  1.3
2  1980  0.600178  f  0.1
```

`get_dummies()` and `from_dummies()`

To convert categorical variables of a `Series` into a “dummy” or “indicator”, `get_dummies()` creates a new `DataFrame` with columns of the unique variables and the values representing the presence of those variables per row.

```
In [60]: df = pd.DataFrame({"key": list("bbacab"), "data1": range(6)})

In [61]: pd.get_dummies(df["key"])
Out[61]:
       a      b      c
0  False   True  False
1  False   True  False
2   True  False  False
3  False  False   True
4   True  False  False
5  False   True  False

In [62]: df["key"].str.get_dummies()
Out[62]:
   a  b  c
0  0  1  0
1  0  1  0
2  1  0  0
3  0  0  1
4  1  0  0
5  0  1  0
```

`prefix` adds a prefix to the column names which is useful for merging the result with the original `DataFrame`:

```
In [63]: dummies = pd.get_dummies(df["key"], prefix="key")

In [64]: dummies
Out[64]:
   key_a  key_b  key_c
0  False   True  False
1  False   True  False
2   True  False  False
3  False  False   True
4   True  False  False
5  False   True  False

In [65]: df[["data1"]].join(dummies)
Out[65]:
   data1  key_a  key_b  key_c
0      0  False   True  False
1      1  False   True  False
2      2   True  False  False
3      3  False  False   True
4      4   True  False  False
5      5  False   True  False
```

This function is often used along with discretization functions like `cut()`:

```
In [66]: values = np.random.randn(10)

In [67]: values
Out[67]:
array([ 0.2742,  0.1329, -0.0237,  2.4102,  1.4505,  0.2061, -0.2519,
       -2.2136,  1.0633,  1.2661])

In [68]: bins = [0, 0.2, 0.4, 0.6, 0.8, 1]

In [69]: pd.get_dummies(pd.cut(values, bins))
Out[69]:
   (0.0, 0.2]  (0.2, 0.4]  (0.4, 0.6]  (0.6, 0.8]  (0.8, 1.0]
0       False        True       False       False       False
1        True       False       False       False       False
2       False       False       False       False       False
3       False       False       False       False       False
4       False       False       False       False       False
5       False        True       False       False       False
6       False       False       False       False       False
7       False       False       False       False       False
8       False       False       False       False       False
9       False       False       False       False       False
```

`get_dummies()` also accepts a `DataFrame`. By default, `object`, `string`, or `categorical` type columns are encoded as dummy variables with other columns unaltered.

```
In [70]: df = pd.DataFrame({"A": ["a", "b", "a"], "B": ["c", "c", "b"], "C": [1, 2,

In [71]: pd.get_dummies(df)
Out[71]:
   C    A_a    A_b    B_b    B_c
0  1   True  False  False   True
1  2  False   True  False   True
2  3   True  False   True  False
```

Specifying the `columns` keyword will encode a column of any type.

```
In [72]: pd.get_dummies(df, columns=["A"])
Out[72]:
   B  C    A_a    A_b
0  c  1   True  False
1  c  2  False   True
2  b  3   True  False
```

As with the `Series` version, you can pass values for the `prefix` and `prefix_sep`. By default the column name is used as the prefix and `_` as the prefix separator. You can specify `prefix` and `prefix_sep` in 3 ways:

- string: Use the same value for `prefix` or `prefix_sep` for each column to be encoded.
- list: Must be the same length as the number of columns being encoded.
- dict: Mapping column name to prefix.

```
In [73]: simple = pd.get_dummies(df, prefix="new_prefix")

In [74]: simple
Out[74]:
   C  new_prefix_a  new_prefix_b  new_prefix_b  new_prefix_c
0  1          True         False         False          True
1  2         False          True         False          True
2  3          True         False          True         False

In [75]: from_list = pd.get_dummies(df, prefix=["from_A", "from_B"])

In [76]: from_list
Out[76]:
   C  from_A_a  from_A_b  from_B_b  from_B_c
0  1      True     False     False      True
1  2     False      True     False      True
2  3      True     False      True     False

In [77]: from_dict = pd.get_dummies(df, prefix={"B": "from_B", "A": "from_A"})

In [78]: from_dict
Out[78]:
   C  from_A_a  from_A_b  from_B_b  from_B_c
0  1      True     False     False      True
1  2     False      True     False      True
2  3      True     False      True     False
```

To avoid collinearity when feeding the result to statistical models, specify `drop_first=True`.

```
In [79]: s = pd.Series(list("abcaa"))

In [80]: pd.get_dummies(s)
Out[80]:
       a      b      c
0   True  False  False
1  False   True  False
2  False  False   True
3   True  False  False
4   True  False  False

In [81]: pd.get_dummies(s, drop_first=True)
Out[81]:
       b      c
0  False  False
1   True  False
2  False   True
3  False  False
4  False  False
```

When a column contains only one level, it will be omitted in the result.

```
In [82]: df = pd.DataFrame({"A": list("aaaaa"), "B": list("ababc")})

In [83]: pd.get_dummies(df)
Out[83]:
    A_a    B_a    B_b    B_c
0  True   True  False  False
1  True  False   True  False
2  True   True  False  False
3  True  False   True  False
4  True  False  False   True

In [84]: pd.get_dummies(df, drop_first=True)
Out[84]:
     B_b    B_c
0  False  False
1   True  False
2  False  False
3   True  False
4  False   True
```

The values can be cast to a different type using the `dtype` argument.

```
In [85]: df = pd.DataFrame({"A": list("abc"), "B": [1.1, 2.2, 3.3]})

In [86]: pd.get_dummies(df, dtype=np.float32).dtypes
Out[86]:
B      float64
A_a    float32
A_b    float32
A_c    float32
dtype: object
```

`from_dummies()` converts the output of `get_dummies()` back into a `Series` of categorical values from indicator values.

```
In [87]: df = pd.DataFrame({"prefix_a": [0, 1, 0], "prefix_b": [1, 0, 1]})

In [88]: df
Out[88]:
   prefix_a  prefix_b
0         0         1
1         1         0
2         0         1

In [89]: pd.from_dummies(df, sep="_")
Out[89]:
  prefix
0      b
1      a
2      b
```

Dummy coded data only requires `k - 1` categories to be included, in this case the last category is the default category. The default category can be modified with `default_category`.

```
In [90]: df = pd.DataFrame({"prefix_a": [0, 1, 0]})

In [91]: df
Out[91]:
   prefix_a
0         0
1         1
2         0

In [92]: pd.from_dummies(df, sep="_", default_category="b")
Out[92]:
  prefix
0      b
1      a
2      b
```

`explode()`

For a `DataFrame` column with nested, list-like values, `explode()` will transform each list-like value to a separate row. The resulting `Index` will be duplicated corresponding to the index label from the original row:

```
In [93]: keys = ["panda1", "panda2", "panda3"]

In [94]: values = [["eats", "shoots"], ["shoots", "leaves"], ["eats", "leaves"]]

In [95]: df = pd.DataFrame({"keys": keys, "values": values})

In [96]: df
Out[96]:
     keys            values
0  panda1    [eats, shoots]
1  panda2  [shoots, leaves]
2  panda3    [eats, leaves]

In [97]: df["values"].explode()
Out[97]:
0      eats
0    shoots
1    shoots
1    leaves
2      eats
2    leaves
Name: values, dtype: str
```

`DataFrame.explode` can also explode the column in the `DataFrame`.

```
In [98]: df.explode("values")
Out[98]:
     keys  values
0  panda1    eats
0  panda1  shoots
1  panda2  shoots
1  panda2  leaves
2  panda3    eats
2  panda3  leaves
```

`Series.explode()` will replace empty lists with a missing value indicator and preserve scalar entries.

```
In [99]: s = pd.Series([[1, 2, 3], "foo", [], ["a", "b"]])

In [100]: s
Out[100]:
0    [1, 2, 3]
1          foo
2           []
3       [a, b]
dtype: object

In [101]: s.explode()
Out[101]:
0      1
0      2
0      3
1    foo
2    NaN
3      a
3      b
dtype: object
```

A comma-separated string value can be split into individual values in a list and then exploded to a new row.

```
In [102]: df = pd.DataFrame([{"var1": "a,b,c", "var2": 1}, {"var1": "d,e,f", "var2"

In [103]: df.assign(var1=df.var1.str.split(",")).explode("var1")
Out[103]:
  var1  var2
0    a     1
0    b     1
0    c     1
1    d     2
1    e     2
1    f     2
```

`crosstab()`

Use `crosstab()` to compute a cross-tabulation of two (or more) factors. By default `crosstab()` computes a frequency table of the factors unless an array of values and an aggregation function are passed.

Any `Series` passed will have their name attributes used unless row or column names for the cross-tabulation are specified

```
In [104]: a = np.array(["foo", "foo", "bar", "bar", "foo", "foo"], dtype=object)

In [105]: b = np.array(["one", "one", "two", "one", "two", "one"], dtype=object)

In [106]: c = np.array(["dull", "dull", "shiny", "dull", "dull", "shiny"], dtype=ob

In [107]: pd.crosstab(a, [b, c], rownames=["a"], colnames=["b", "c"])
Out[107]:
b    one        two
c   dull shiny dull shiny
a
bar    1     0    0     1
foo    2     1    1     0
```

If `crosstab()` receives only two `Series`, it will provide a frequency table.

```
In [108]: df = pd.DataFrame(
   .....:     {"A": [1, 2, 2, 2, 2], "B": [3, 3, 4, 4, 4], "C": [1, 1, np.nan, 1, 1
   .....: )
   .....:

In [109]: df
Out[109]:
   A  B    C
0  1  3  1.0
1  2  3  1.0
2  2  4  NaN
3  2  4  1.0
4  2  4  1.0

In [110]: pd.crosstab(df["A"], df["B"])
Out[110]:
B  3  4
A
1  1  0
2  1  3
```

`crosstab()` can also summarize to `Categorical` data.

```
In [111]: foo = pd.Categorical(["a", "b"], categories=["a", "b", "c"])

In [112]: bar = pd.Categorical(["d", "e"], categories=["d", "e", "f"])

In [113]: pd.crosstab(foo, bar)
Out[113]:
col_0  d  e
row_0
a      1  0
b      0  1
```

For `Categorical` data, to include **all** of data categories even if the actual data does not contain any instances of a particular category, use `dropna=False`.

```
In [114]: pd.crosstab(foo, bar, dropna=False)
Out[114]:
col_0  d  e  f
row_0
a      1  0  0
b      0  1  0
c      0  0  0
```

Normalization

Frequency tables can also be normalized to show percentages rather than counts using the `normalize` argument:

```
In [115]: pd.crosstab(df["A"], df["B"], normalize=True)
Out[115]:
B    3    4
A
1  0.2  0.0
2  0.2  0.6
```

`normalize` can also normalize values within each row or within each column:

```
In [116]: pd.crosstab(df["A"], df["B"], normalize="columns")
Out[116]:
B    3    4
A
1  0.5  0.0
2  0.5  1.0
```

`crosstab()` can also accept a third `Series` and an aggregation function (`aggfunc`) that will be applied to the values of the third `Series` within each group defined by the first two `Series`:

```
In [117]: pd.crosstab(df["A"], df["B"], values=df["C"], aggfunc="sum")
Out[117]:
B    3    4
A
1  1.0  NaN
2  1.0  2.0
```

Adding margins

`margins=True` will add a row and column with an `All` label with partial group aggregates across the categories on the rows and columns:

```
In [118]: pd.crosstab(
   .....:     df["A"], df["B"], values=df["C"], aggfunc="sum", normalize=True, marg
   .....: )
   .....:
Out[118]:
B       3    4   All
A
1    0.25  0.0  0.25
2    0.25  0.5  0.75
All  0.50  0.5  1.00
```

`cut()`

The `cut()` function computes groupings for the values of the input array and is often used to transform continuous variables to discrete or categorical variables:

An integer `bins` will form equal-width bins.

```
In [119]: ages = np.array([10, 15, 13, 12, 23, 25, 28, 59, 60])

In [120]: pd.cut(ages, bins=3)
Out[120]:
[(9.95, 26.667], (9.95, 26.667], (9.95, 26.667], (9.95, 26.667], (9.95, 26.667], (9
Categories (3, interval[float64, right]): [(9.95, 26.667] < (26.667, 43.333] < (43.
```

A list of ordered bin edges will assign an interval for each variable.

```
In [121]: pd.cut(ages, bins=[0, 18, 35, 70])
Out[121]:
[(0, 18], (0, 18], (0, 18], (0, 18], (18, 35], (18, 35], (18, 35], (35, 70], (35, 7
Categories (3, interval[int64, right]): [(0, 18] < (18, 35] < (35, 70]]
```

If the `bins` keyword is an `IntervalIndex`, then these will be used to bin the passed data.

```
In [122]: pd.cut(ages, bins=pd.IntervalIndex.from_breaks([0, 40, 70]))
Out[122]:
[(0, 40], (0, 40], (0, 40], (0, 40], (0, 40], (0, 40], (0, 40], (40, 70], (40, 70]]
Categories (2, interval[int64, right]): [(0, 40] < (40, 70]]
```

`factorize()`

`factorize()` encodes 1 dimensional values into integer labels. Missing values are encoded as `-1`.

```
In [123]: x = pd.Series(["A", "A", np.nan, "B", 3.14, np.inf])

In [124]: x
Out[124]:
0       A
1       A
2     NaN
3       B
4    3.14
5     inf
dtype: object

In [125]: labels, uniques = pd.factorize(x)

In [126]: labels
Out[126]: array([ 0,  0, -1,  1,  2,  3])

In [127]: uniques
Out[127]: Index(['A', 'B', 3.14, inf], dtype='object')
```

`Categorical` will similarly encode 1 dimensional values for further categorical operations

```
In [128]: pd.Categorical(x)
Out[128]:
['A', 'A', NaN, 'B', 3.14, inf]
Categories (4, object): [3.14, inf, 'A', 'B']
```

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